

Title: **Electric Vehicles vs. Charger Availability**

Problem Statement: Is the lack of charger availability a major contributor to the reason why the general population is reluctant about making the switch to driving an electric vehicle?

Additional Questions:

- In states where there are more chargers available, are there more registered owners of electric vehicles? (and vice versa)
- Is there an even distribution of chargers across the U.S? (This is important for road trips or anyone who may drive long distances at some point during the ownership of their vehicle.)
- Do charger speeds appear to have any overall impact? (For example, too many slow chargers outside of major cities leading to extended charging times in rural areas.)
- If charging stations were as available as gas stations, would most people prefer to drive an electric vehicle?

Background: This is a significant problem due to global warming and air pollution. With millions of drivers on the road each day, the emissions are starting to impact our environment. These issues have resulted in the state of California aiming to only sell zero emission vehicles by 2035. As a result, other states have also followed along by implementing more electric vehicle charging stations.

Potential: Our project has the potential to contribute to our problem by addressing the main issue when it comes to making the switch to an electric vehicle which is, how to power it? We as a society are so used to the availability of gas stations while electric vehicle chargers seem few and far between. Our project will ultimately determine if the number of chargers is directly correlated with the number of people willing to own an electric vehicle. This contribution is crucial because if they are correlated, that means we as a country must work towards making charging stations as available as gas stations to ensure a smooth transition to electric vehicles and help the environment.

Data Cleaning/Processing:

Cleaning step 1:

```
# cleaning step 1

# Drop columns with mostly null values or unrelated to electric charging
cols_to_drop = [
    'Plus4', 'Expected Date', 'BD Blends', 'NG Fill Type Code', 'NG PSI',
    'Hydrogen Status Link', 'NG Vehicle Class', 'LPG Primary', 'E85 Blender Pump',
    'Intersection Directions (French)', 'Access Days Time (French)', 'BD Blends (French)',
    'Groups With Access Code (French)', 'CNG Dispenser Num', 'CNG On-Site Renewable Source',
    'CNG Total Compression Capacity', 'CNG Storage Capacity', 'LNG On-Site Renewable Source',
    'EV Pricing (French)', 'LPG Nozzle Types', 'Hydrogen Pressures', 'Hydrogen Standards',
    'CNG Fill Type Code', 'CNG PSI', 'CNG Vehicle Class', 'LNG Vehicle Class',
    'RD Blends', 'RD Blends (French)', 'RD Blended with Biodiesel',
    'RD Maximum Biodiesel Level', 'CNG Station Sells Renewable Natural Gas', 'Intersection Directions',
    'Station Phone', 'Cards Accepted', 'EV Other Info', 'Geocode Status', 'Owner Type Code',
    'Federal Agency ID', 'Federal Agency Name', 'Open Date', 'Hydrogen Is Retail',
    'Access Code', 'Access Detail Code', 'Federal Agency Code', 'Facility Type',
    'E85 Other Ethanol Blends', 'EV Pricing', 'EV On-Site Renewable Source',
    'Restricted Access', 'NPS Unit Name', 'LNG Station Sells Renewable Natural Gas',
    'Maximum Vehicle Class', 'EV Workplace Charging', 'EV Network Web',
    'Access Days Time'
]

df.drop(columns=cols_to_drop, inplace=True, errors='ignore')
```

This first step in our cleaning process simply removes a ton of columns that are either mostly null values or offer no real contribution in answering our overall question. If there were fewer rows, I would offer a more precise explanation on each removal but I can guarantee that they are either mostly null and or offer no significance. Once this step was completed I was left with only the raw data that directly contributes to what our project is trying to answer.

Cleaning step 2:

```
# cleaning step 2

df.rename(columns={
    'EV Level1 EVSE Num': 'Level1_Chargers',
    'EV Level2 EVSE Num': 'Level2_Chargers',
    'EV DC Fast Count': 'DC_Fast_Chargers',
    'EV Network': 'EV_Network',
    'EV Connector Types': 'Connector_Types'
}, inplace=True)
```

Cleaning step two simply renames a few columns in order to make them more readable / understandable. Additionally, this will make our graphs more understandable later on in the project.

Cleaning step 3:

```
# cleaning step 3  
  
# only keep rows where the state is a two letter abbreviation  
df = df[df['State'].notnull() & df['State'].str.match("^[A-Z]{2}$")]
```

Cleaning step 3 ensures that the state column only contains state names as two letter abbreviations. This is important for when we merge our two datasets together later.

Cleaning step 4:

```
# cleaning step 4  
  
# Convert Charger Counts to Numeric  
df['Level1_Chargers'] = pd.to_numeric(df['Level1_Chargers'], errors='coerce')  
df['Level2_Chargers'] = pd.to_numeric(df['Level2_Chargers'], errors='coerce')  
df['DC_Fast_Chargers'] = pd.to_numeric(df['DC_Fast_Chargers'], errors='coerce')
```

Cleaning step four converts our charger count variables from a string to numeric. This is crucial for when we implement them into a graph later on for analysis.

Cleaning step 5:

```
# cleaning step 5

# Remove Duplicate Stations
df.drop_duplicates(subset=['Station Name', 'Street Address', 'City', 'State'], inplace=True)
```

Cleaning step 5 checks for and removes all duplicate charging stations. This ensures that our data is not skewed by false information.

Cleaning step 6:

```
# cleaning step 6

# Drop rows missing long / lat locations
df = df[df['Latitude'].notnull() & df['Longitude'].notnull()]
```

Cleaning step 6 removes all rows that don't have latitude and longitude locations. This is important to ensure that our data is only represented by chargers with coordinates. By doing this we can accurately compare our graphs which results in a more accurate answer to our overall question. Including data without a precise location may skew our data in a negative way.

Cleaning step 7:

```
# cleaning step 7

# fill missing charger counts with 0
df[['Level1_Chargers', 'Level2_Chargers', 'DC_Fast_Chargers']] = df[['Level1_Chargers', 'Level2_Chargers', 'DC_Fast_Chargers']].fillna(0)
```

Cleaning step removes all null values in charger counts with 0. This ensures we get all data on the charging station while leaving our accurate charging count data unaffected. The type of chargers available is a question we're looking to answer in addition to our main goal and the few stations with no counts have minimal impact on our findings.

Cleaning step 8:

```
# Cleaning step 8  
  
# create total chargers column  
df['Total_Chargers'] = df['Level1_Chargers'] + df['Level2_Chargers'] + df['DC_Fast_Chargers']
```

Cleaning step 8 adds a brand new column at the end of our dataset containing the total number of chargers in the specified station. We will use this data to compute the total number of chargers in each state and then compare that to the number of registered electric vehicles.

Cleaning step 9:

```
# cleaning step 9  
  
# Standardize cases in names  
df['EV_Network'] = df['EV_Network'].str.title()  
df['City'] = df['City'].str.title()
```

Cleaning step 9 ensures that all of the city and EV_Network names follow the same formatting in terms of which letters are uppercase. This standard makes our data cleaner, more readable, and makes graphs look more presentable.

Cleaning step 10:

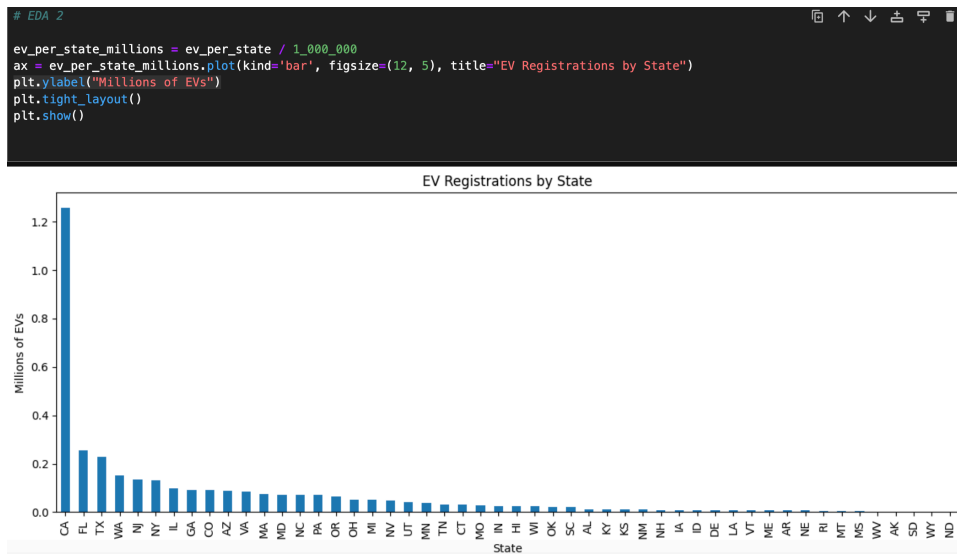
```
# cleaning step 10  
  
# Keep only electric vehicle charging stations  
# dataset includes multiple types of alternative fuel stations  
df = df[df["Fuel Type Code"] == "ELEC"]
```

Cleaning step 10 ensures that all of the chargers in our database are 100% electric. This was done because it's apparent that the U.S is slowly moving its way to an all electric platform so it's important that our data only reflects charging stations that are powered with electricity.

```
# EDA 1

df.info()
df.describe()
print("\nmissing values\n")
df.isnull().sum()
```

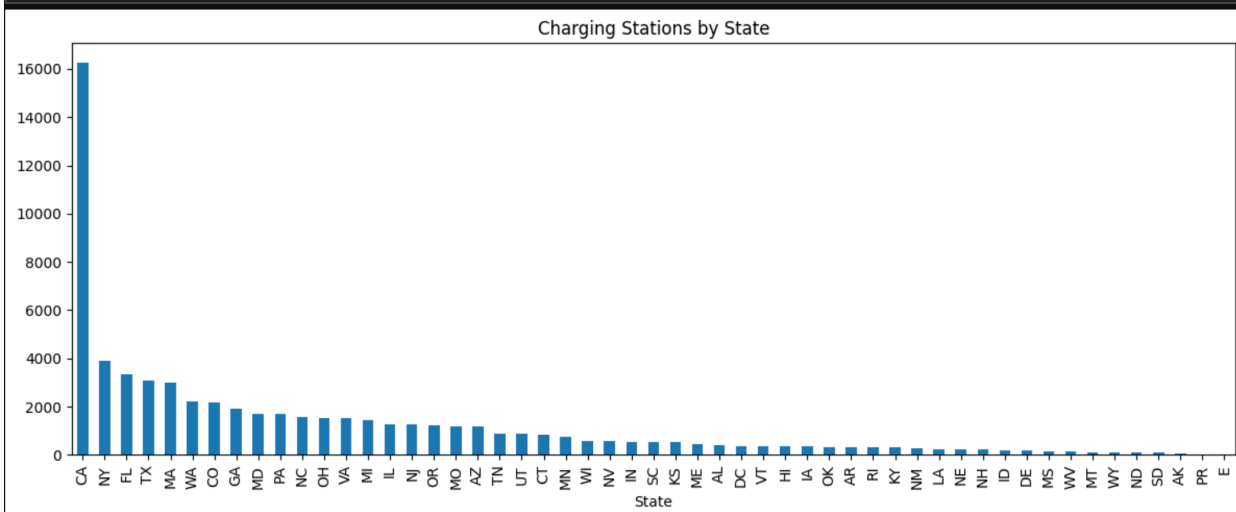
EDA operation 1 is very simple. It provides an overview of our now cleaned and merged dataset and gives us a handful of analytics on our data. This allows us to confirm all previous steps were done correctly and also provides us with data that can be used later on when creating graphs. For example, knowing which data type is being used for a column.



EDA operation 2 creates a bar graph displaying the number of registered electric vehicles by state. We learned that the large majority of electric vehicles are located in CA, FL, and TX. We will use this graph to compare with others and draw new conclusions.

```
# EDA 3
```

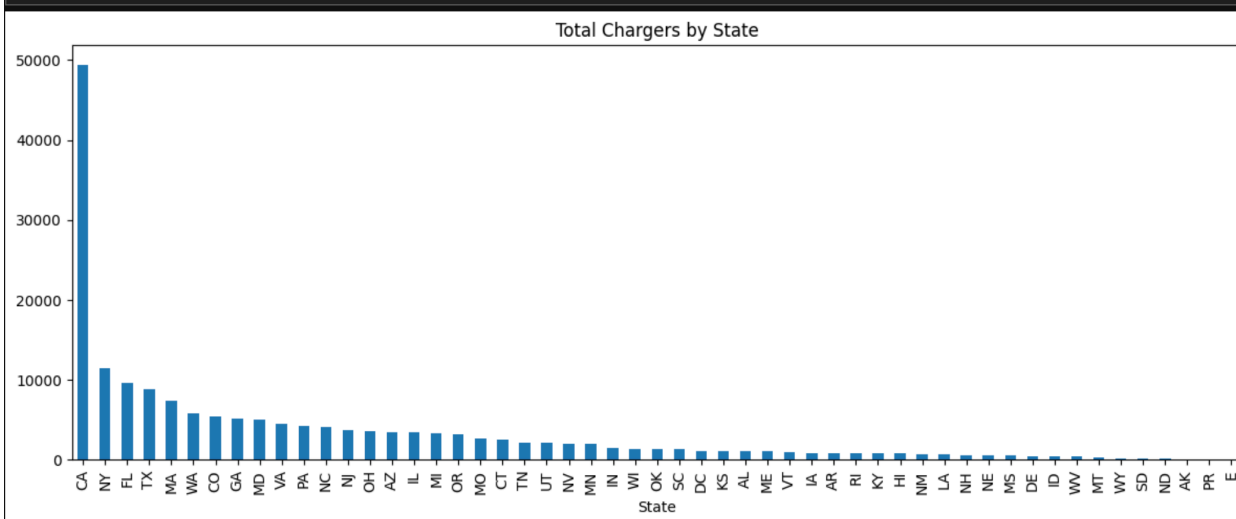
```
stations_per_state = df.groupby('State').size().sort_values(ascending=False)
stations_per_state.plot(kind='bar', figsize=(12, 5), title="Charging Stations by State")
plt.tight_layout()
plt.show()
```



EDA operation 3 creates a bar graph showing the number of charging stations by state. We can use this graph and compare it to our previous graph of registrations. We learned at this point that the number of charging stations seems to be correlated to the number registered vehicles in each state. However, we had more analysis to do.

```
# EDA 4
```

```
chargers_per_state = df.groupby('State')['Total_Chargers'].sum().sort_values(ascending=False)
chargers_per_state.plot(kind='bar', figsize=(12, 5), title="Total Chargers by State")
plt.tight_layout()
plt.show()
```

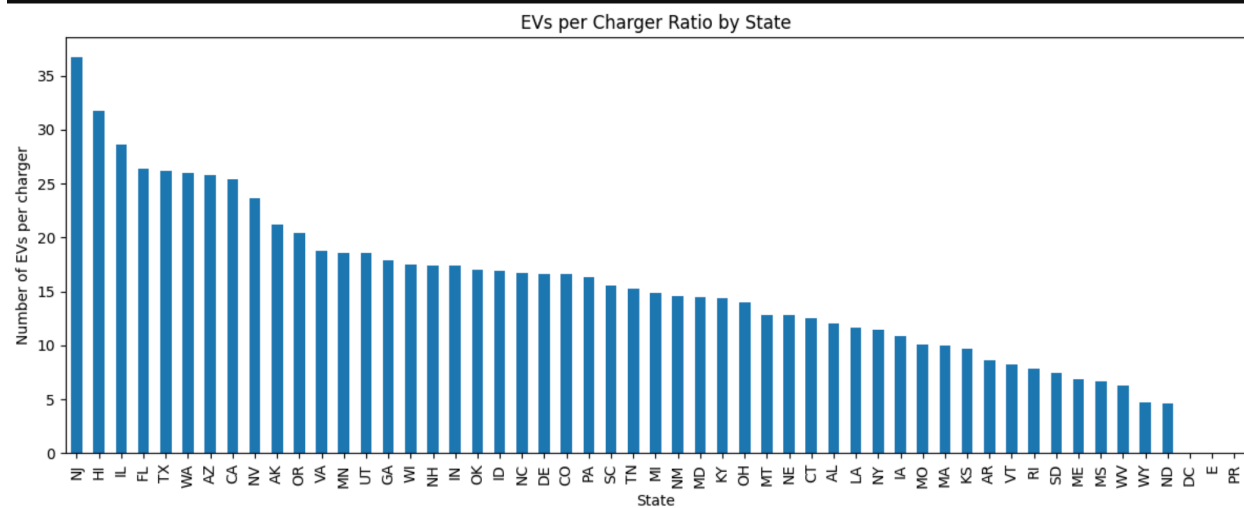


EDA operation 4 creates a bar graph of the number of individual chargers in each state. This allows us to see if it's simply the number of charging stations that matter or the number of

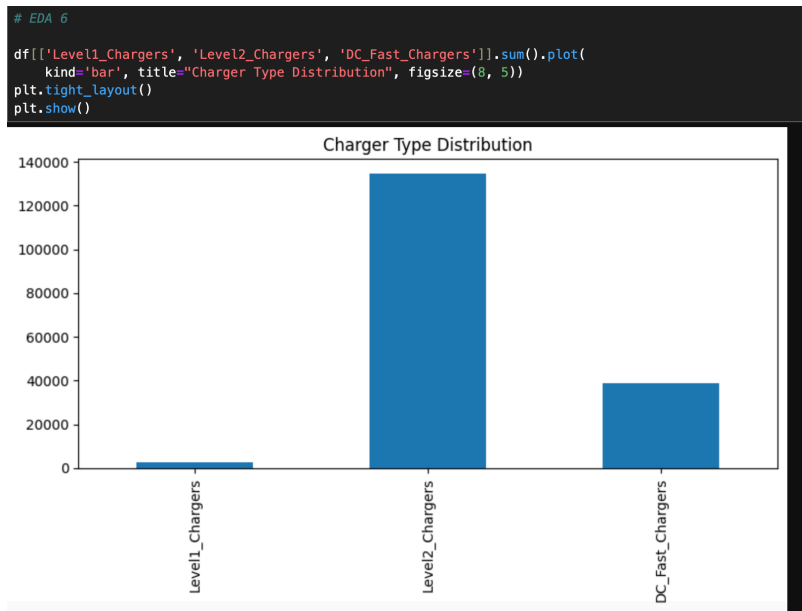
chargers specifically. However, we learned that the charging stations and the number of chargers are mostly correlated.

```
# EDA 5

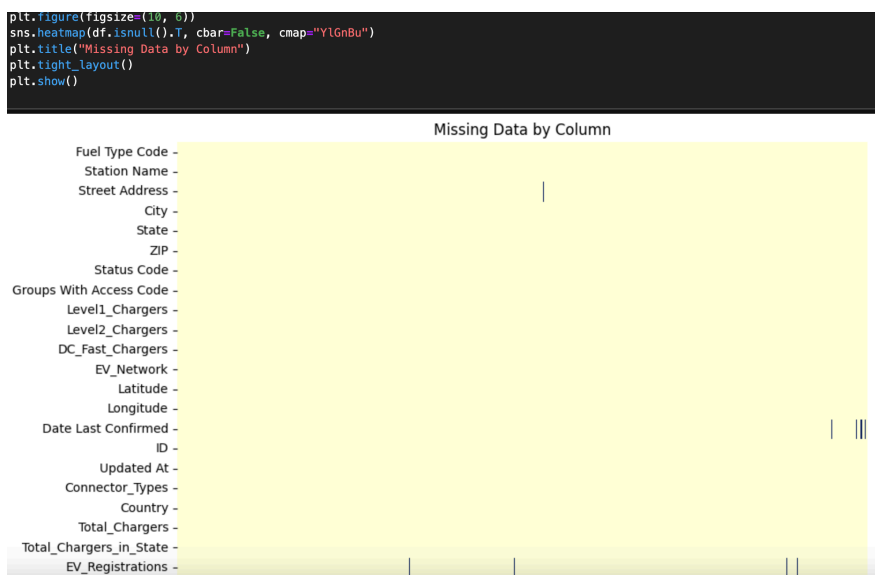
evs = state_df.groupby('State')['EV_Registrations'].first()
chargers = df.groupby('State')['Total_Chargers'].sum()
ev_charger_ratio = (evs / chargers).sort_values(ascending=False)
ev_charger_ratio.plot(kind='bar', figsize=(12, 5), title="EVs per Charger Ratio by State")
plt.ylabel("Number of EVs per charger")
plt.tight_layout()
plt.show()
```



EDA operation 5 creates a bar graph showing the number of registered vehicles per every one charger available. For example, there are 35 registered electric vehicles in New Jersey for each charger that exists in the state. We learned that New Jersey is most likely one of the worst places to own an electric vehicle due to charger availability.

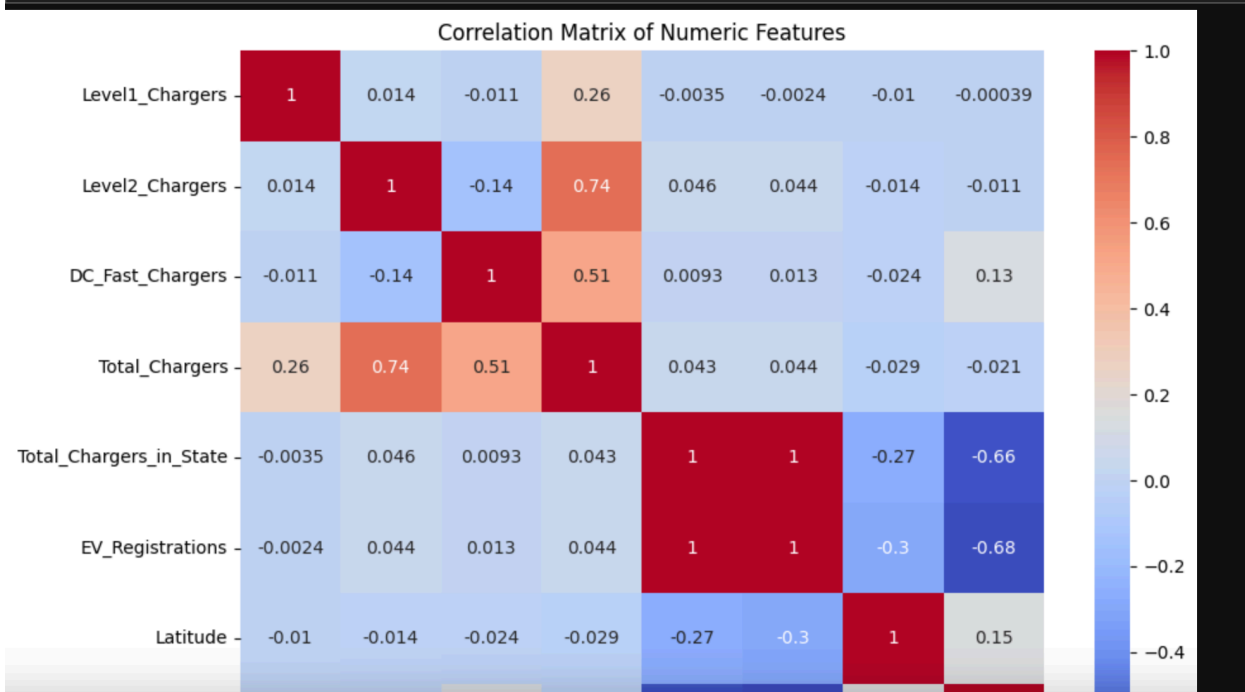


EDA operation 6 creates a bar graph that shows the amount of each type of charger in the whole country. Through this graph we learned that the large majority of chargers in the US are level 2 chargers. We also learned that the type of charger available likely has little to no effect on whether or not someone would want to own an electric vehicle.



(Run program for full picture) EDA Operation 7 creates a table that gives us a visual representation of where our missing data is in our dataset. The yellow represents non missing data while the green and blue lines represent where missing data is. We learned that our dataset is clean and our missing values don't have a significant impact on our conclusions we made earlier.

```
plt.figure(figsize=(10, 8))
sns.heatmap(df[numeric_cols].corr(), annot=True, cmap="coolwarm")
plt.title("Correlation Matrix of Numeric Features")
plt.tight_layout()
plt.show()
```



(Run program to view full photo) EDA operation 8 creates a correlation matrix that compares our numerical columns to see if they are correlated. The most notable piece of information that we learned from creating this table is that the number of chargers in each state is directly correlated to the number of registered electric vehicles.

EDA 9

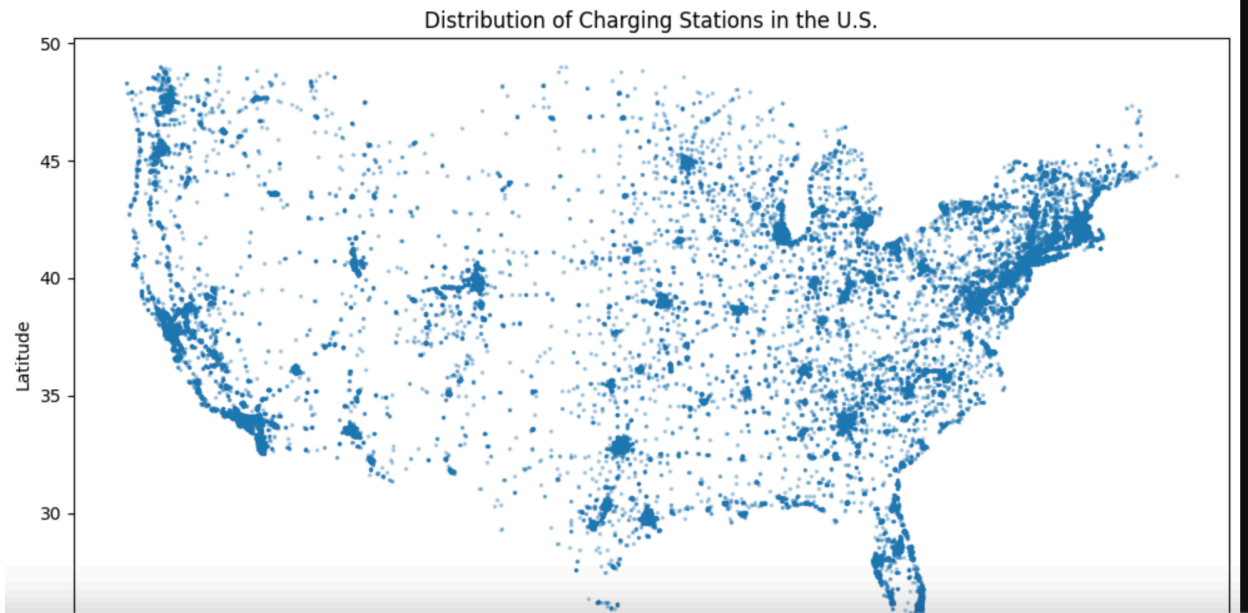
```
outliers = ev_charger_ratio[ev_charger_ratio > ev_charger_ratio.quantile(0.95)]
print("Worst 5% of EV to Charger Ratios:\n", outliers.head())
```

Worst 5% of EV to Charger Ratios:

```
State
NJ    36.707437
HI    31.718362
IL    28.637619
dtype: float64
```

EDA operation 9 provides us with the worst 5% of states to own an electric vehicle when it comes to the ratio of registered electric vehicles per every 1 charger. Through this data we learned that NJ, HI, and IL may be the most difficult states to find an available charger at.

```
plt.figure(figsize=(10, 6))
plt.scatter(df_clean['Longitude'], df_clean['Latitude'], alpha=0.3, s=2)
plt.title("Distribution of Charging Stations in the U.S.")
plt.xlabel("Longitude")
plt.ylabel("Latitude")
plt.tight_layout()
plt.show()
```



EDA operation 10 provides a map of all the charging stations in the US. Through this data we learned that the north east has the most charging stations distributed over a large area. Where as CA, TX, and FL have the most stations in a more confined space around major cities.